MIL-W-17265/3A(SHIPS)
18 January 1968
SUPERSEDING
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(See 6.2)

MILITARY SPECIFICATION

WINCH, DRUM, POWER OPERATED

(INHAUL WINCH TYPE 8.0-IH-2D AND OUTHAUL WINCH TYPE 8.0-OH-2D)

1. SCOPE

1.1 Scope. This specification covers the detail requirements for inhaul and outhaul winches.

1.2 Classification. Winches shall be classified in accordance with the following types as specified (see 6.1).

Type 8.0-IH-2D - Double drum inhaul winch rated 8,000 pounds at 250 feet per minute and 4000 pounds at 500 feet per minute.

Type 8.0-OH-2D - Double drum outhaul winch rated 8,000 pounds at 250 feet per minute and 4000 pounds at 500 feet per minute.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or requests for proposal, form a part of the specification to the extent specified herein.

SPECIFICATIONS

FEDERAL
RR-W-410 - Wire Rope and Strand.

MILITARY
MIL-W-17265 - Winch, Drum, Power Operated (Electric-Hydraulic, variable speed)

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 General requirements. Winches, drum, power operated, type 8.0-IH-2D and type 8.0-OH-2D shall be in accordance with MIL-W-17265, except as specified herein.

3.2 Drums. Drums for type 8.0-IH-2D and 8.0-OH-2D Winches shall be designed for the following:

(a) Drum provided with controlled spooling system.
  (1) Stowage capacity for 1,200 feet of 1/2 inch wire rope. The breaking strength of the wire rope is 25,200 pounds.
  (2) Barrel diameter of 24 inches plus or minus 1/2 inch.
  (3) Distance between flanges shall be 15 inches plus or minus 1/2 inch.
(b) Drum not provided with controlled spooling system.
  (1) Stowage capacity for 600 feed of 3/4 inch wire rope. The breaking strength of the wire rope is 48,600 pounds.
  (2) Barrel diameter of 24 inches plus or minus 1/2 inch.
  (3) Distance between flanges shall be 15 inches plus or minus 1/2 inch.

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3.2.1 Wire rope construction. Wire rope specified for the winch drums with controlled spooling shall be Type I, Class 3, 6 x 37 construction, independent wire core (IWRC), extra improved plow steel, uncoated, preformed, right regular lay in accordance with RR-W-410.

3.2.2 Controlled spooling. - The drum on type 8.0-IH-2D winch used for the inhaul and the drum on type 8.0-OH-2D used for the outhaul shall be provided with controlled spooling. The controlled spooling shall insure that the wire spools on the drum in an evenly spaced, interrupted spiral pattern from flange to flange at all times when the equipment is in operation as well as during start up, after periods of disuse without making adjustments. The first and each succeeding layer of wire on the drum shall form a spooling pattern for the next layer that will exactly reproduce this spooling pattern on all layers. The spooling pattern of wire on the drum shall eliminate overriding (cross-over) turns and undercutting of the wire rope.

3.2.3 This type of controlled spooling shall exist during all periods of winch operation when the winch is used for tensioned highline transfer and during the rigging and unrigging phases of the tensioned highline rig. The requirements of 3.2.3.1 and 3.2.3.2 shall also be complied with.

3.2.3.1 The controlled spooling system shall operate at zero line pull and maximum winch speeds in both haul-in and payout directions.

3.2.3.2 The controlled spooling system shall operate at maximum winch deceleration and maximum acceleration between full speed haul-in and full speed payout for all winch loads from zero to rated load.

3.2.4 Wire rope life requirements. The winch controlled spooling system shall provide at least 24,000 cycles (see 3.2.4.1) of operation without failure (see 3.2.4.3) of the wire rope. Ten percent of the 24,000 cycles shall be conducted at zero line pull under maximum speed change between full speed haul-in and full speed payout. Ninety percent of the 24,000 cycles shall be conducted at winch rated load and rated speed at the specified cycle rate (see 3.2.4.2).

3.2.4.1 Cycle. A cycle is defined as passage of a section of wire rope from the winch drum through the winch controlled spooling system as well as any components of the winch system the wire rope normally runs over and back onto the winch drum.

3.2.4.2 Cycle rate. Cycle rate when operating winch rated speed and rated load is defined as one cycle every fifteen cycles.

3.2.4.3 Wire rope failure. Wire rope failure is defined as seven or more broken wires in one strand in one rope lay.

3.2.5 Controls. The winch controlled spooling system shall be provided with controls as necessary to permit initial spooling of wire rope on the winch and through the shipboard underway replenishment system. If controls for the shipboard underway replenishment operation including passing and returning the rig are required, they shall be located in the underway control station.

3.3 Modes of operation. Winch types specified in 1.2 shall be provided with speed, tension and position control modes of operation.

3.3.1 Speed control modes of operation. The rated line pull and line speed for speed mode of operation shall be as follows:

(a) Drum provided with controlled spooling system.
  (1) Rated line pull - 4000 pounds on the average layer of the drum.
  (2) Rated line speed - 500 feet per minute.
(b) Drum not provided with controlled spooling system.
  (1) Rated line pull - 8,000 pounds on the average layer of the drum.
  (2) Rated line speed - 250 feet per minute.

3.3.2 Automatic tension mode of operation.

3.3.2.1 Reel in speed. The automatic tension control of the winch shall be designed so that reeling speed will vary inversely as a function of tension. That is, the speed of the winch shall automatically increase as tension decreases.

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3.3.2.2 Tension level. The automatic tension control of the winch shall be designed so that it will cause the winch to payout or haul-in to maintain required tension level.

3.3.2.3 Speed tension values. The speed-tension values at maximum tension setting shall follow the curve formed by points A, B and C. The speed at point A is 900 feet per minute (payout) and at point C the speed is 900 feet per minute (haul-in). Curve BC shall be a constant power curve formed by the following equation:

Speed (fpm haul-in) times Tension (pounds) equals $2x10^6$.

The curve from point A to B shall be a straight line and shall have a tension decrease of one pound for each foot per minute speed change; shall pass through 5500 pounds at zero feet per minute, to intersection with curve BC.

3.3.2.4 Reduced tension setting. At reduced tension setting the curve shall parallel AB until the speed reaches 900 feet per minute or intersects curve BC in which case it will follow curve BC to point C.

3.3.2.4.1 Minimum tension setting shall pass through 1000 pounds at zero feet per minute.

3.3.3 Overhauling load. The winch design shall permit the winch to be overhauled by an external force at speeds from 900 to 1400 feet per minute. The tension value during this period may be any value required to protect winch components from damage, including power generation, provided the tension does not exceed 6900 pounds.

3.3.4 Automatic recovery. The winch design shall provide automatic recovery from 900 to 1400 feet per minute overhauling speed condition back to normal tension control range (900 feet minute payout to 900 feet per minute haul-in).

3.3.5 Adjustments. The design of the automatic tension control shall be such that reference points A through C could be modified either through adjustments or substitution of components.

3.3.6 Actual tension values. Actual tension values shall be within ± 100 pounds of the values specified in 3.3.2.3 above.

3.3.7 Instantaneous cable tension. The tension control shall prevent the cable tension from exceeding 8000 pounds when an instantaneous cable tension increase occurs as would be experienced when the load on the cable is two blocked against a fixed stop at 900 feet per minute with not more than thirty feet of wire are between the winch and the fixed stop.

3.3.8 Tension accuracy. The winch shall provide tension accuracy at all tension settings of ± 10 percent of the rated tension during acceleration and deceleration at rates up to 6.7 feet per second squared.

3.3.9 Rate of tension increase. The rate at which the tension level can be raised from minimum to maximum values shall be restricted to a minimum of two seconds.

3.3.10 Position control mode of operation. Position control mode of operation shall provide the following winch operation in response to ship separation and rate of change signals obtained from the high line under control of the ram tensioner:

(a) When the trolley is at the supply ship; (1) hold the inhaul winch stationary, (2) cause the outhaul winch to control a working length of wire rope equal to twice the working length of the high line between ships, to payout or haul in wire rope at twice the speed, footage, and acceleration and in the same direction that the high line is moving in response to relative ship motion.
(b) When trolley is at the receiving ship, cause both the inhaul and the outhaul winch to control a working length of rope equal to the working length of the high line between ships, to payout or haul in wire rope at the same speed, footage, and acceleration and in the same direction that the highline is moving in response to relative ship motion.
(c) With the trolley at the supply ship, upon signal from the operator, discontinue the type of control described in (a) above; cause the trolley to accelerate and move at maximum available speed; while still compensating for relative ship motion, travel to a point outboard of the receiving ship; decelerate; stop, and assume the type of control described in (b) above.

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(d) With the trolley at the receiving ship, upon signal from the operator, discontinue the type of control described in (b) above; cause the trolley to accelerate and move at maximum available speed; while still compensating for relative ship motion, travel to a point outboard of the supply ship; decelerate; stop, and assume the type of control described in (a) above.
(e) Provide both a minimum and a maximum tension override for function's (a), (b), (c), & (d) above.
(f) Provide means of interrupting functions (c) and (d) above stopping and holding the trolley between ships and then continuing the transit operation or returning the trolley to its starting point upon command from the operator.
(g) Depending upon transfer system being used, the control for (a) and (b) above shall be suitable for systems using trolley catching and locking devices on both ships; systems using no mechanical stop on either ship or systems using any combination of the above.
(h) When no mechanical stop is used the winch operator shall be able to adjust the stop point over a range of 1 to 20 feet with an accuracy of plus or minus one foot, at either ship.

3.4 Duty. Winch types specified in 1.2 shall be designed to permit continuous operation for a period of twelve hours for the normal operating cycle specified below. A mornal operating cycle will consist of 12 seconds at rated speed rated load; 18 seconds at idle, 12 seconds at half rated speed and rated load; and 18 seconds at idle.

3.4.1 Name plate ampere rating of the electric motor shall not be exceeded when the winch is being operated at rated load and rated speed (constant velocity), temperature rise shall not exceed name plate permissible temperature rise when the winch is operated at the duty cycle and period specified in 3.8.

3.4.2 Thermal overload. Thermal overload shall be indicated by energizing a master panel indicating light through a thermal heating device (see thermal requirements of MIL-W-17265) in either the motor or the controller. The indicator light shall be in accordance with the illumination requirements of MIL-W-17265. This device shall not de-energize the motor.

3.5 Torque clutch. Each drum provided with controlled spooling shall be equipped with a torque (slip) clutch. This clutch shall be the driving member between the drum shaft and the drum. The tension (slip) value specified below for the various winch types are based on the wire being on the average layer of the drum. The clutch shall be designed so as not to use any external energy source and shall be self compensating for normal wear.

3.5.1 The torque clutch for the winch types specified in 1.2 shall slip at a maximum of 8,000 pounds and reset at a minimum of 7,000 pounds.

3.6 Hydraulic brake. When in speed control mode of operation the hydraulic brake shall automatically set upon loss of servo pressure. When in automatic tension and position control mode of operation the hydraulic back shall be designed to stay in the released position for at least 15 minutes upon loss of servo pressure.

3.7 Relief valve. The high pressure relief valve (hoist side) for the hydraulic system shall automatically vent to a lower pressure value, to limit the maximum load than can be imposed on the wire rope, upon loss of electric power to the winch.

3.7.1 The lower pressure value for the relief valve shall be adjustable and shall be set at a value that will permit the winch to be overhauled with a rope pull of 900 to 1,000 pounds (wire rope on the average layer) when operating in either the tension or position mode of operation. This requirement is based on the hydraulic brake released (see 3.10).

3.8 Hydraulic motors. The hydraulic motor for winch types specified in 1.2 shall be of the variable delivery type. A position indicator shall be provided to indicate the setting of the motor. In the event of loss of power to the system the hydraulic motor shall go to minimum stroke.

3.9 Size and Weight. The winch shall be as compact and light as is consistent with adequate strength and rigidity. Weight and size shall not exceed the following:

(a) Maximum weight - 15,000 pounds
(b) Maximum width (across drum) with gypsy head installed - 100 inches
(c) Maximum length - 120 inches
(d) Maximum height - 96 inches.

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3.10 Methods of Operation.

3.10.1 Required methods of operation shall be in accordance with the Master speed control unit requirements, and the winch speed control unit requirements of MIL-W-17265, and the following:

(a) Position control (see para 3.10.2)
(b) Tension control (see para 3.10.3)

3.10.2 Master position control.

3.10.2.1 A position control and it's associated equipment, capable of performing the functions specified in 3.3.10, shall be added to the master control panel.

3.10.2.2 A position shall be added to the operations selector switch in accordance with the secondary remote control unit requirements of MIL-W-17265.

3.10.3 Master Tension Control.

3.10.3.1 A master tension control shall be added to the operations selector switch in accordance with the secondary remote control unit requirements of MIL-W-17265.

3.10.3.2 Associated controls required to perform the functions specified in 3.3.2 shall also be added to the master control panel.

3.10.3.3 The lever for the master speed function shall control the tension operation as follows:

(a) When the operations selector switch is turned counterclockwise to the master tension position, the lever arm, in its spring centered position shall establish minimum tension in the system.
(b) Pulling the lever arm towards the operator shall increase tension to maximum.
(c) Returning the lever arm forward to the center position shall decrease tension to minimum.
(d) The system shall remain in minimum tension for all lever positions forward of center position.

3.10.3.4 A placard shall be installed next to the lever arm showing direction and function of the master speed and master tension methods of operation.

3.10.4 Indicator light. An indicator light shall be added to the panel for each method of operation. The lights and the dimmer control shall be in accordance with the illumination requirements of MIL-W-17265.

3.10.5 Replenishment at sea control stations. Replenishment at sea control stations shall be of an integrated console design in lieu of miscellaneous arrangement of control and indicating lights.

3.10.5.1 Master control panel. A master control panel shall consist of all controls and indicating lights for the inhaul and outhaul winches. Controls shall be symmetrical about the vertical center line. The lever arm for the outhaul winch shall be on the left hand side and the lever arm for the inhaul winch shall be on the right side of the panel.

3.11 Gypsy heads. One gypsy head shall be installed for the winch types specified in 1.2.

3.12 Gearing. Gear drives shall have a service factor of 1.25.

3.13 Sheaves. All sheaves shall be designed with a miminum sheave tread diameter to wire rope diameter ratio of 30 to 1.

4. QUALITY ASSURANCE PROVISIONS

4.1 Tests. In addition to the tests specified in MIL-W-17265, the tests specified in 4.2 through 4.4 shall be conducted.

4.2 Torque clutch. Torque clutch tests shall be made at the manufacturer's plant.

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4.2.1 Each drum torque clutch shall be tested by pulling a 80 foot length of wire off of the drum at a speed of not less than 700 feet per minute. The maximum tension measured during this test shall be recorded. (The use of a special locking device on the winch to hold the drum shaft stationary during this test is acceptable.)

4.2.2 The torque clutch of one winch of each type in the lot shall be tested by repeating the above test two times, waiting approximately 15 seconds between runs, and seven additional times, allowing the clutch to cool as necessary between runs. The maximum tension measured during each run shall be recorded.

4.3 Automatic tension and position control modes of operation. One set of inhaul and outhaul winches offered for delivery on the contract or order shall be tested for conformance to 3.3.2 through 3.3.10.

4.4 Controlled spooling system and wire rope life tests. One winch of each class offered for delivery one one contract or order shall be tested to show conformance with all requirements specified for the controlled spooling system (see 3.2.2 through 3.2.4).

4.6 Procedures. The procedures for conducting tests specified in 4.2 through 4.4 shall be submitted to the Naval Ship Engineering Center (NAVSEC) Weapons, Deck Systems, and Equipment Branch, Department of the Navy, Washington, D. C. 20360, for approval 60 days prior to schedule dates for testing.

4.7 Record of tests. A copy of data recorded in conducting the tests required by 4.2 through 4.4 shall be forwarded to NAVSEC within thirty days after completion of tests.

5. PREPARATION FOR DELIVERY

5.1 Preparation for delivery shall be in accordance with MIL-W-17265.

6. NOTES

6.1 Ordering data. Ordering data shall be as specified in section 6 of MIL-W-17265.

6.2 CHANGES FROM PREVIOUS ISSUE. THE OUTSIDE MARGINS OF THIS DOCUMENT HAVE BEEN MARKED "#" TO INDICATE WHERE CHANGES (DELETIONS, ADDITIONS, ETC.) FROM THE PREVIOUS ISSUE HAVE BEEN MADE. THIS HAS BEEN DONE AS A CONVENIENCE ONLY AND THE GOVERNMENT ASSUMES NO LIABILITY WHATSOEVER FOR ANY INACCURACIES IN THESE NOTATIONS. BIDDERS AND CONTRACTORS ARE CAUTIONED TO EVALUATE THE REQUIREMENTS OF THIS DOCUMENT BASED ON THE ENTIRE CONTENT AS WRITTEN IRRESPECTIVE OF THE MARGINAL NOTATIONS AND RELATIONSHIP TO THE LAST PREVIOUS ISSUE.

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